
Venkat Neelraj Nitta

Boston, MA | [Portfolio](#) | vneelraj.nitta@gmail.com | +1(857)-693-2870 |

[LinkedIn:v-neelraj-nitta](#) | [Git:Subhash-269](#) |

Summary

Aspiring Software/AI Engineer passionate about building intelligent **automations, dashboards**, and troubleshooting tools that enhance service efficiency. Experienced in integrating **AI models, APIs**, and **cloud workflows** on **AWS** and **Azure** to drive operational and analytical improvements. Proficient in **Python, SQL**, and **data pipelines** with a solid foundation in **backend systems, vector databases**, and **LLM frameworks**. Available **Spring 2026** for a **full-time co-op**.

Education

Northeastern University, Boston, MA September 2024 – Expected April 2026
Master of Professional Studies in Applied Machine Intelligence, STEM **GPA: 3.96**

Related courses: Fundamentals of Artificial Intelligence, Applications of AI, AI System Technologies (ML Ops), Healthcare Information Processing, Healthcare/Pharmaceutical Data and Applications.

Mahindra Ecole Centrale, Hyderabad, India July 2022
Bachelor of Engineering in Electrical Electronics Engineering

Related courses: Signals and Systems, Data Structures, Numerical Methods, Statistics, Digital Signal Processing, Machine Learning, Natural Language Processing, Advance Data Analytics, Big Data Computing.

Technical Skills

Programming and Frameworks	Python, SQL, PyTorch, Django, FastAPI, Flask, Docker, Linux.
ML and CV	Classical ML, Evaluation Metrics, Image Classification, OCR, Computer Vision, Object Detection, Predictive Modeling.
Data & MLOps	Pandas, NumPy, Feature Engineering, Data Pipelines, CI/CD, MLflow, AWS (S3, Lambda, SageMaker), Azure ML.
LLM & NLP Tools	LangChain, LangGraph, Hugging Face, RAG Pipelines, OpenAI API, Vector Databases (FAISS, Pinecone).

Professional Experience

Software/AI Developer, Cognida.ai – Hyderabad, India July 2022 – August 2024

- Engineered **backend architecture** and **APIs** for an **AI-powered inspection platform** integrating OCR and computer vision; implemented **dashboards and data pipelines**, improving processing speed **by 90%** and reducing costs **by 60%**.
- Led **LLM fine-tuning** and **evaluation** across cloud platforms (**AWS, Azure**); built **RAG-based chatbot workflows** using embeddings and vector search, reducing query resolution time **from 10 days to minutes**.
- Developed a **healthcare claims analytics system** to flag insurance claims with potential third-party recovery opportunities; built **data pipelines** and predictive features to enhance model **accuracy** and **automation**.

Software/AI Developer Intern, Cognida.ai – Hyderabad, India February 2022 – June 2022

- Developed and optimized an **NLP-based language translation and grammar correction model**, ensuring context-aware, accurate corrections with **real-time performance**.
- Enhanced model reliability through **feedback mechanisms** and an **iterative retraining workflow**.

Hackathons

InnovAIt, Northeastern: Smart Cart App

February 2025

Github: [Smart-Cart](#)

Tools: Django, PostgreSQL, React, Google Gemini API, Hugging Face

- Led the development of **SmartCart**, an app that provides competitive pricing by automatically identifying the **lowest vendor prices** and reducing grocery shopping time.
- Built and deployed a **pricing analytics and decision-support prototype**, improving **user efficiency** and enabling **real-time insights**.

Cognida Cognida Competition, Cognida.ai: Object Detection Web Application

February 2022

- Developed a **PyTorch-based object detection system**, overcoming limited training data through **augmentation and data scraping**, achieving the **highest precision** among participants.
- Delivered a working **computer vision inspection pipeline** and secured an internship based on model **accuracy, speed, and deployment quality**.

Projects

Automobile Insurance Assistance

October 2025

GitHub: <https://github.com/Subhash-269/InsurAsst>

Skills: Python, LangChain, YOLOv11, RAG pipelines, Teamwork, Collaboration, Curiosity

- Built a **multimodal GenAI system** for insurance claim guidance using **YOLOv11** and agentic frameworks (**LangChain, LangGraph**); enabled users to query policy documents or upload vehicle images for **damage detection and claim assistance**.
- Deployed a **Django web app** supporting real-time **vector search, embeddings, and context-aware decision tracking** for insurance claims.

CIFAR-10 Image Classification Web App

February 2024

GitHub: github.com/Subhash-269/cifar-image-cls-app

Live App: cifar-image-classifier.streamlit.app

Technologies: Python, PyTorch, FastAPI, Streamlit

- **Fine-tuned** a ResNet-18 model, applied model optimization (quantization + ONNX) for low-latency inference.
- Built a **full-stack ML application** with a FastAPI backend and Streamlit-based real-time visualization system, integrating an SQLite logging pipeline to support image uploads, prediction tracking, and a continuous model feedback loop.

Certifications

- **MLOps; Machine Learning in Production, DeepLearning.AI**

November 2025

- **Generative AI with Large Language Models, DeepLearning.AI**

May 2024

- **Applying AI Technologies to the Workplace, Northeastern University**

Septembere 2024

- **Foundations of Modern Machine Learning, IIIT-Hyderabad**

July 2022

- **Intro to Machine Learning Course, Kaggle**

2021

- **Intermediate Machine Learning Course, Kaggle**

2021